NDSS PAPER UPDATE

RV-MalTrace

Hardware-assisted behavior tracing for RISC-V Linux workloads

Scope: controlled Genesys2/CVA6 tracing and semantic reconstruction. Not a malware detector or production transport claim.

TRACE EVENTS FIRST

`artifact root: results/evaluation/genesys2-cva6/current`

Software-only tracing changes the observation surface

Software tracers are useful references, but visible to workloads.

SOFTWARE OBSERVER

Rich semantics

Useful reference data, but visible to the workload.

OUT-OF-BAND TRACE

Hardware path

Commit-time trace starts below user-mode observer checks.

Trace events first; derived semantics carry provenance

The paper claims a trace-backed behavior analysis pipeline.

TRACE EVENTS with **PROVENANCE**

Committed events are hardware observations.

Provenance means source tags for derived labels.

Three contributions define the paper

Each contribution maps to code and evidence in the repository.

C1

CVA6 trace tap

Out-of-band commit, privilege, and drop events.

C2

Semantic reconstruction

Derived behavior labels with provenance.

C3

Validation artifacts

Genesys2/CVA6 runs, manifests, and checkers.

The application is trace-backed behavior analysis

Hardware trace, local code analysis, and malware-behavior rules are joined.

HARDWARE TRACE

Trace events

syscalls, traps, returns, drops

LOCAL CODE

Attribution

ELF hash, symbols, syscall sites

MALWARE ANALYSIS

Behavior rules

rule families, lineage, limitations

Core claim: trace-backed behavior analysis, not a mature classifier.

Implementation joins three data sources

Trace events stay separate from code attribution and behavior labels.

SEPARATE	hardware trace is kept separate from reference labels
TAG	code and behavior claims carry provenance

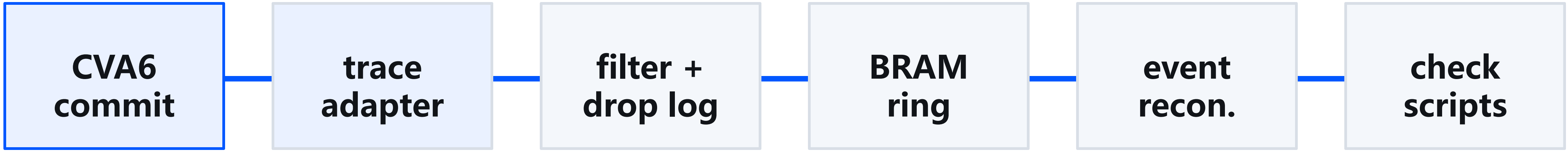
Implemented as a scoped prototype

35T is end-to-end; Genesys2/CVA6 supports scoped evidence.

35T	13/13 matrix PASS; 6/6 behavior cases PASS
CVA6	12 case studies PASS; 8 safe surrogates
LIMIT	no accuracy, payload-equivalence, or source-line claim

CVA6 commit events feed semantic reconstruction

The CVA6 path is sideband trace plus offline reconstruction.



Trace records capture transitions and drops

The trace format is the contract between RTL and analysis.

CONTROL	retire, branch, jump
SYSCALL	entry and qualified return
CONTEXT	trap, CSR, SATP, privilege
ACCOUNT	ARG_MEM, marker, drop records

Code maps make trace evidence attributable

ELF identity, symbols, syscall sites, and runtime maps constrain attribution.

CODE MAP

ELF identity

hash, load ranges, sections, symbols

JOIN

Trace to code

PC evidence plus process/runtime scope

LIMITATION

No source-line claim

DWARF/source-line attribution needs board evidence

The 35T end-to-end path is implemented

35T connects trace capture, code-map join, and behavior rules.

13/13
synthetic matrix

512
trace records

PASS
fd/path + process tree

6/6
malware-derived behavior cases

Implemented as a safety-controlled prototype; not malware-family accuracy.

CVA6 supports the current paper evidence

It is not yet a real-malware validation or source-line attribution claim.

PASS	12 controlled case studies in current manifest
PASS	8 safe malware-behavior surrogate case studies
LIMITED	local code fixtures pass; source-line attribution not claimed
NO CLAIM	real malware validation and detector accuracy

Separate supported, limited, and unsupported claims

The paper must separate implementation evidence from nonclaims.

SUPPORTED

Core evidence

correctness, provenance, board runs, resource/timing

LIMITED

Scoped claims

source lines, strings, benign controls, tracer baseline

UNSUPPORTED

Not claimed

transport, overhead, real malware validation

122 accepted board runs

This is the strongest current CVA6 empirical result.

122

accepted board runs

12
samples

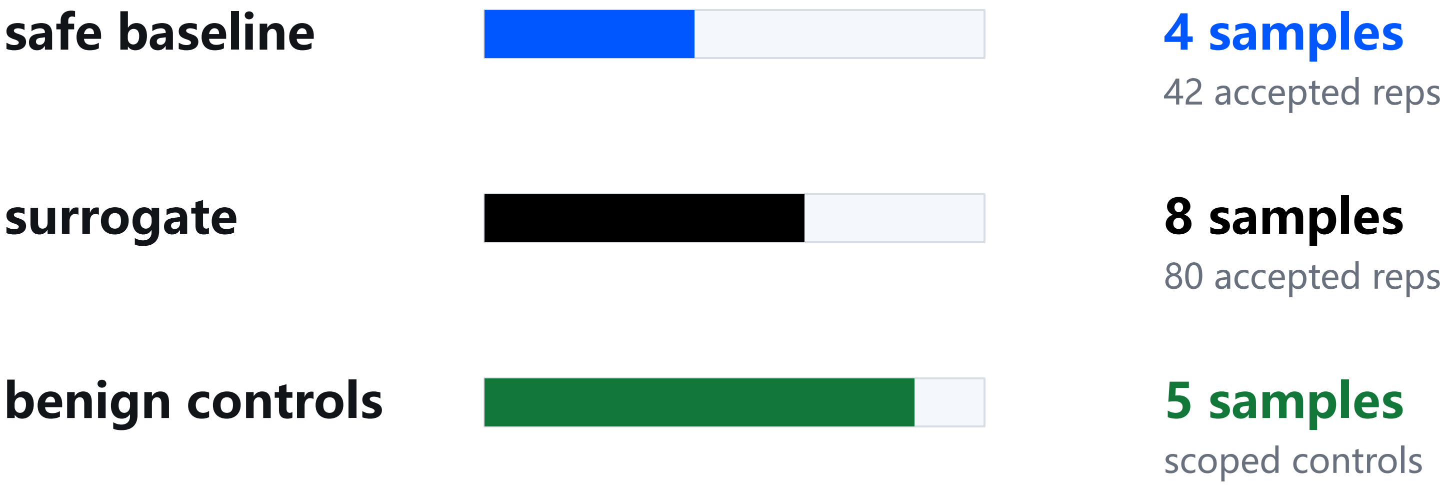
126
attempts

4
failed retained

0
drops / wraps / gaps

Controlled workloads, not in-the-wild malware

The workload set supports tracing validation under controlled behavior coverage.



Rule-based behavior analysis passes scoped checks

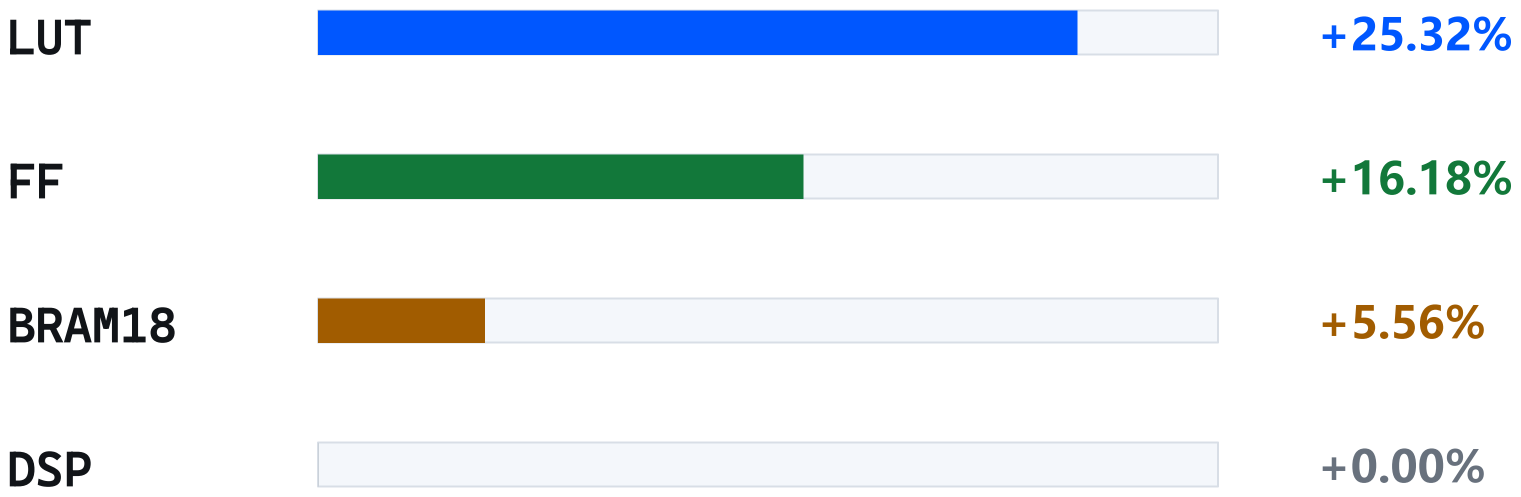
These metrics validate reconstruction, not detector accuracy.

1.0	syscall recall
1.0	precision
1.0	argument recon.
0.0	benign FPR

Controlled workload analysis only. Not detection accuracy.

Resource cost is measured; overhead is not

Timing/resource evidence supports implementation cost, not cycle slowdown.



Timing MET, slack 0.177 ns. Cycle-level runtime overhead remains unsupported.

The artifact package is reviewable

Evidence packaging and checkers are ahead of manuscript completeness.

MANIFEST	current artifact root accepted
ARCHIVE	3160 files, 316.9 MB, SHA d5b806...433b
CHECKERS	repro:quick and documented full gates pass
PAPER	LaTeX skeleton builds, content incomplete

Claim boundaries are explicit

Unsupported topics become limitations or future work.

WRITE	hardware-assisted behavior tracing and analysis
WRITE	35T end-to-end prototype; scoped CVA6 evidence
LIMIT	synthetic and safety-controlled malware-derived behaviors
DO NOT	real-malware accuracy, payload equivalence, production detector

Paper narrative

Hardware trace + code attribution + rule-based malware-behavior analysis.

Problem

visible software observers

Mechanism

commit events + code map

Application

rule-based analysis

Boundary

no mature detector claim

Streaming/DMA remains future work

Readiness and target profiles are not throughput evidence.

TARGET	p99 = 0.0215755 event bytes/cycle
REQUIRED	0.0323633 event bytes/cycle, about 1.618 MB/s at 50 MHz
CURRENT	sustained streaming = 0; noninterference = false

Workload roster

Sample names are appendix material, not main-story material.

SAFE	hello_write; file_open_read_write; fork_exec; illegal_instruction
SURROGATE	file_scan; batch_open_read_write; self_copy_sim; abnormal_syscall_sequence
SURROGATE	illegal_trap; process_chain; dynamic_executable_memory; anti_debug_like
BENIGN	hello; cat; clock_status; getdents_only; mmap_rw

Resource detail

Resource/timing claim is supported; runtime overhead is not.

	LUT	FF	BRAM18	DSP
baseline	84,923	56,491	108	27
trace	106,428	65,634	114	27
delta	+25.32%	+16.18%	+5.56%	+0